CLAIMS

1. A method of screening for cancer in a subject, said method comprising determining the level and/or chemical composition of the protein-free fraction of one or both of the glycosaminoglycans (GAGs) chondroitin sulfate (CS) and heparan sulfate (HS) in a body fluid sample, wherein said sample has been obtained from said subject and has been subjected to processing prior to determining said level and/or chemical composition,

wherein said processing

(a) comprises fragmenting said one or both GAGs into disaccharide units; and

(b) does not comprise prior to (a):

(i) contacting said sample with a proteolytic agent; or

(ii) purifying said one or both GAGs in said sample based on the negative charge of said GAGs.

2. The method of claim 1, wherein an altered level and/or chemical composition of chondroitin sulfate (CS) and/or heparan sulfate (HS) in said protein-free fraction in comparison to a control level and/or chemical composition is indicative of cancer in said subject.

3. The method of claim 1 or claim 2,

wherein said determination of the chemical composition comprises determining the level in said protein-free fraction of one or more GAG properties selected from the group consisting of: one or more (or all) of the specific sulfated or unsulfated forms of CS or HS disaccharides, charge HS, charge CS, the total concentration of CS or the total concentration of HS, and/or

wherein said determination of the chemical composition comprises determining the level in said protein-free fraction of one or more (or all) of the GAG properties selected from the group consisting of: the specific sulfated or unsulfated forms of CS or HS disaccharides selected from the group consisting of: 0s CS, 2s CS, 6s CS, 4s CS, 2s6s CS, 2s4s CS, 4s6s CS,Tris CS, 0s HS, 2s HS, 6s HS, 2s6s HS, Ns HS, Ns2s HS, Ns6s HS, Tris HS, the ratio of 4s CS to 6s CS, the ratio of 6s CS to 0s CS

and the ratio of 4s CS to 0s CS; charge HS; charge CS; the total concentration of CS (CS tot); and the total concentration of HS (HS tot).

4. The method of any one of claims 1 to 3, wherein said body fluid sample is blood (e.g. plasma) and/or urine.

5. The method of any one of claims 1 to 4, wherein said determination of the chemical composition comprises determining the level in said protein-free fraction of

(i) one or more (or all) of the GAG properties selected from the group consisting of: the specific sulfated or unsulfated forms of CS or HS disaccharides selected from the group consisting of: the specific sulfated or unsulfated forms of CS or HS disaccharides selected from the group consisting of: absolute concentration of 4s CS, absolute concentration of 0s CS, relative concentration of 0s CS, absolute concentration of 0s HS, absolute concentration of Ns HS, absolute concentration of 6s CS, absolute concentration of 2s6s CS, relative concentration of 4s CS, relative concentration of 2s6s CS, relative concentration of 6s CS, relative concentration of 0s HS, relative concentration of Ns HS; the ratio of 6s CS to 0s CS; the ratio of 4s CS to 0s CS; CS tot; HS tot; charge CS; or

(ii) one or more (or all) of the GAG properties selected from the group consisting of: the specific sulfated or unsulfated forms of CS or HS disaccharides selected from the group consisting of: absolute concentration of 4s CS, absolute concentration of 0s CS, relative concentration of 0s CS, absolute concentration of 0s HS, absolute concentration of Ns HS, absolute concentration of 6s CS, absolute concentration of 2s6s CS, relative concentration of 4s CS, relative concentration of 2s6s CS, relative concentration of 6s CS; CS tot; HS tot; charge CS; or

(iii) one or more (or all) of the GAG properties selected from the group consisting of: the specific sulfated or unsulfated forms of CS or HS disaccharides selected from the group consisting of: absolute concentration of 4s CS, absolute concentration of 0s CS, relative concentration of 0s CS, absolute concentration of 0s HS, absolute concentration of Ns HS, absolute concentration of 2s6s CS, relative concentration of 4s CS, relative concentration of 2s6s CS, relative concentration of 6s CS, relative concentration of 0s HS, relative concentration of Ns HS; the ratio of 6s CS to 0s CS; the ratio of 4s CS to 0s CS; CS tot; HS tot; charge CS; or

(iv) one or more (or all) of the GAG properties selected from the group consisting of: the specific sulfated or unsulfated forms of CS or HS disaccharides selected from the group consisting of: absolute concentration of 4s CS, absolute concentration of 0s CS, relative concentration of 0s CS, absolute concentration of 0s HS, absolute concentration of Ns HS, absolute concentration of 2s6s CS, relative concentration of 4s CS, relative concentration of 2s6s CS, relative concentration of 6s CS, relative concentration of 0s HS, relative concentration of Ns HS, absolute concentration of 6s CS; the ratio of 6s CS to 0s CS; the ratio of 4s CS to 0s CS; CS tot; HS tot; charge CS; or

(v) one or more (or all) of the GAG properties selected from the group consisting of: the specific sulfated or unsulfated forms of CS or HS disaccharides selected from the group consisting of: absolute concentration of 0s CS, relative concentration of 0s CS, absolute concentration of 4s CS, relative concentration of 4s CS; the ratio of 4s CS to 0s CS; CS Tot; charge CS in a blood (e.g. plasma) sample; or

(vi) one or more (or all) of the GAG properties selected from the group consisting of: the specific sulfated or unsulfated forms of CS or HS disaccharides selected from the group consisting of: absolute concentration of 0s CS, relative concentration of 0s CS, absolute concentration of 4s CS, relative concentration of 4s CS; CS Tot; charge CS in a blood (e.g. plasma) sample; or

(vii) one or more (or all) of the GAG properties selected from the group consisting of: the specific sulfated or unsulfated forms of CS or HS disaccharides selected from the group consisting of: absolute concentration of 0s CS, relative concentration of 4s CS; the ratio of 4s CS to 0s CS in a blood (e.g. plasma) sample; or

(viii) one or more (or all) of the GAG properties selected from the group consisting of: the specific sulfated or unsulfated forms of CS or HS disaccharides selected from the group consisting of: absolute concentration of 4s CS, absolute concentration of 0s CS, relative concentration of 0s CS, absolute concentration of 0s HS, absolute concentration of Ns HS, absolute concentration of 6s CS, absolute concentration of 2s6s CS, relative concentration of 4s CS, relative concentration of 2s6s CS, relative concentration of 6s CS, relative concentration of 0s HS, relative concentration of Ns HS; the ratio of 6s CS to 0s CS; CS tot; HS tot; charge CS in a urine sample; or

(ix) one or more (or all) of the GAG properties selected from the group consisting of: the specific sulfated or unsulfated forms of CS or HS disaccharides selected from the group consisting of: absolute concentration of 4s CS, absolute

concentration of 0s CS, relative concentration of 0s CS, absolute concentration of 0s HS, absolute concentration of Ns HS, absolute concentration of 6s CS, absolute concentration of 2s6s CS, relative concentration of 2s6s CS, relative concentration of 6s CS; CS tot; HS tot; charge CS in a urine sample; or

(x) one or more (or all) of the GAG properties selected from the group consisting of: the specific sulfated or unsulfated forms of CS or HS disaccharides selected from the group consisting of: absolute concentration of 4s CS, relative concentration of 0s CS, absolute concentration of 0s HS, absolute concentration of Ns HS, absolute concentration of 2s6s CS, relative concentration of 4s CS, relative concentration of 2s6s CS, relative concentration of 6s CS, relative concentration of 0s HS, relative concentration of Ns HS; the ratio of 6s CS to 0s CS; CS tot; HS tot in a urine sample; or

(xi) one or more (or all) of the GAG properties selected from the group consisting of: the specific sulfated or unsulfated forms of CS or HS disaccharides selected from the group consisting of: absolute concentration of 0s HS, relative concentration of 6s CS, relative concentration of Ns HS, relative concentration of 0s HS, relative concentration of 2s6s CS, relative concentration of 4s CS, relative concentration of 0s CS, absolute concentration of 4s CS, absolute concentration of 6s CS, absolute concentration of 0s CS; the ratio of 6s CS to 0s CS; CS tot; HS tot in a urine sample.

6. The method of any one of claims 1 to 4, wherein

(i) the chemical composition of said one or both of said GAGs in said protein-free fraction of a blood (e.g. plasma) sample is determined and the chemical composition of said one or both of said GAGs in said protein-free fraction of a urine sample is performed, preferably

wherein determination of the chemical composition of said protein-free fraction of said urine sample comprises determining the level in said protein-free fraction of one or more (or all) of the GAG properties selected from the group consisting of: relative concentration of 0s CS, absolute concentration of 0s HS, absolute concentration of Ns HS, relative concentration of 2s6s CS, relative concentration of 0s HS, relative concentration of Ns HS; HS tot; and charge CS; and

wherein determination of the chemical composition of said protein-free fraction of said blood (e.g. plasma) sample comprises determining the level in said protein-free

fraction of one or more (or all) of the GAG properties selected from the group consisting of: absolute concentration of 0s CS; relative concentration of 4s CS; and CS tot; or

(ii) the chemical composition of said one or both of said GAGs in said protein-free fraction of a blood (e.g. plasma) sample is determined and the chemical composition of said one or both of said GAGs in said protein-free fraction of a urine sample is performed,

wherein determination of the chemical composition of said protein-free fraction of said urine sample comprises determining the level in said protein-free fraction of one or more (or all) of the GAG properties selected from the group consisting of: absolute concentration of 0s HS, relative concentration of Ns HS, relative concentration of 0s CS, relative concentration of 0s HS, absolute concentration of 4s CS, absolute concentration of 0s CS, relative concentration of 4s CS, relative concentration of 6s CS, absolute concentration of 6s CS; the ratio of 6s CS to 0s CS; CS tot; and charge CS; and

wherein determination of the chemical composition of said protein-free fraction of said blood (e.g. plasma) sample comprises determining the level in said protein-free fraction of one or both of the GAG properties selected from the group consisting of: the ratio of 4s CS to 0s CS; and CS tot; or

(iii) the chemical composition of said one or both of said GAGs in said protein-free fraction of a blood (e.g. plasma) sample is determined and the chemical composition of said one or both of said GAGs in said protein-free fraction of a urine sample is performed,

wherein determination of the chemical composition of said protein-free fraction of said urine sample comprises determining the level in said protein-free fraction of one or more (or all) of the GAG properties selected from the group consisting of: absolute concentration of 0s HS, absolute concentration of Ns HS, absolute concentration of 4s CS, absolute concentration of 6s CS; and

wherein determination of the chemical composition of said protein-free fraction of said blood (e.g. plasma) sample comprises determining the level in said protein-free fraction of absolute concentration of 0s CS.

7. The method of any one of claims 1 to 6,

wherein determination of the level and/or chemical composition of said protein-free fraction of said body fluid sample comprises determining the level in said protein-free fraction of one or both of the GAG properties selected from the group consisting of: the absolute concentration of 0s CS and CS Tot; and/or

wherein said method comprises determining the level in said protein-free fraction of the absolute concentration of 0s CS and wherein an increase in the absolute concentration of 0s CS, in comparison to a control level, is indicative of cancer.

8. The method of any of claims 3 to 7, wherein the chemical composition may be expressed in terms of score, said score being based on the measured level of one or more (preferably more than one) or all of said GAG properties.

9. The method of any one of claims 1 to 8, wherein said method comprises determining the level of more than one of said GAG properties, preferably said method comprises determining the level of two or more, three or more, four or more, or all, of said GAG properties.

~~10. The method of any one of claims 1 to 9, wherein~~

~~(i) said sample has been obtained from said subject and has been subjected to at least one processing step prior to determining said level and/or chemical composition, preferably wherein said at least one processing step does not comprise contacting said sample with a proteolytic agent; and/or~~

~~(ii) said method is a method of any one of claims 3 to 18, wherein the levels of one or more of the specific sulfated or unsulfated forms of CS or HS disaccharides are determined, and wherein the GAGs are, or have been, subjected to a processing step to obtain the disaccharide units for analysis, preferably wherein said at least one processing step does not comprise contacting said sample with a proteolytic agent.~~

~~11. The method of any one of claims 1 to 10, wherein said sample has been obtained from said subject and has been subjected to processing prior to determining said level and/or chemical composition,~~

~~wherein said processing~~

~~(a) comprises fragmenting said one or both GAGs into disaccharide units, preferably said fragmenting of (a) is performed by contacting said one or both GAGs with one or more GAG lyase enzymes such as one or more chondrotinase enzymes and/or one or more heparinase enzymes; and~~

~~(b) does not comprise prior to (a) at least one of:~~

~~(i) contacting said sample with a proteolytic agent; and~~

~~(ii) purifying said one or both GAGs in said sample based on the negative charge of said GAGs (e.g. using an anion exchange resin),~~

~~preferably said method does not comprise the contacting of (b)(i) or the purifying of (b)(ii).~~

10. The method of any one of claims 1 to 9, wherein said fragmenting of (a) is performed by contacting said one or both GAGs with one or more GAG lyase enzymes.

11. The methof of claim 10, wherein said one or more GAG lyase enzymes are one or more chondrotinase enzymes and/or one or more heparinase enzymes.

12. The method of any one of claims 1 to 11, wherein said level or chemical composition of said GAG or GAG property is determined by HPLC and mass spectrometry, preferably wherein said HPLC is ultra-HPLC and/or said mass spectrometry is triple-quadropole mass spectrometry.

13. The method of any one of claims 1 to 12,

wherein said subject is a subject at risk of developing cancer, or at risk of the occurrence of cancer, or is a subject having or suspected of having cancer; and/or

wherein said cancer is

(i) a stage I cancer, a stage II cancer, a stage III cancer, a stage IV cancer and a cancer of an unspecified stage; or

(ii) selected from the group consisting of a genitourinary cancer (e.g. kidney cancer, prostate cancer or bladder cancer), a respiratory tract cancer (e.g. lung cancer), a brain tumor, a blood cancer (e.g. a lymphoma), colorectal cancer, uterine

cancer, a gastrointestinal-neuroendocrine tumour, a breast cancer, ovarian cancer and head and neck cancer; or

(iii) selected from the group consisting of: bladder cancer, breast invasive ductal carcinoma, cervix squamous cell carcinoma, chronic lymphoid leukaemia, colorectal cancer, endometrial cancer, diffuse glioma, a gastro-intestinal endocrine tumour, head and neck squamous cell carcinoma, diffuse large B-cell lymphoma, non-small cell lung cancer, ovarian cancer, prostate cancer and renal cell cancer.

14. The method of any one of claims 1 to 13, wherein said method is used for diagnosing cancer, for the prognosis of cancer, for predicting the occurrence of cancer, for estimate the risk of the occurrence of cancer, for monitoring subjects at risk of the occurrence of cancer, for monitoring the progression of cancer in a subject, for determining the clinical severity of cancer, for predicting the response of a subject to therapy or surgery for cancer, for determining the efficacy of a therapeutic or surgical regime being used to treat cancer, for detecting the recurrence of cancer, for predicting the tissue of origin of a cancer, or for distinguishing small masses suspicious of cancer from other non malignant diseases.

15. A method of determining the level and/or chemical composition of one or both of the glycosaminoglycans (GAGs) chondroitin sulfate (CS) and heparan sulfate (HS) in a body fluid sample,

wherein said sample has been obtained from said subject and has been subjected to processing prior to determining said level and/or chemical composition,

wherein said processing

(a) comprises fragmenting said one or both GAGs into disaccharide units; and

(b) does not comprise prior to (a) ~~at least one of~~:

(i) contacting said sample with a proteolytic agent;

~~and~~or

(ii) purifying said one or both GAGs in said sample based on the negative charge of said GAGs.